

For testing auto timer cut-off circuit, put sensor A' and D into water. The bulb must glow and remain on for 30 minutes before switching off.

After successful completion of the above tests, the controller is ready for connecting to the submersible pump starter unit as shown in Fig. 5.

Connection diagram of controller to pump starter unit

Fig. 5 shows the connection diagram of water level controller unit to pump starter unit. The step-by-step connection procedure is as follows:

1. Switch off mains supply.
2. Disconnect the 230V AC input wires to the starter unit and connect them to the controller cabinet input points 1 and 2. Connect phase and neutral accordingly.
3. Connect controller output points

3 and 4 to the input terminals of the pump starter unit.

4. The pump-starter green button has normally four points—two for capacitor and two for contactor. Connect terminal points 5 and 6 of the controller to the capacitor points, and points 7 and 8 to the contactor points of the starter button.

Connect sensor probes of OHT with the help of four-wire telephone wire to the respective points (9, 10, 11 and 12) of the terminal block of controller unit.

EFY note. Check that all wires are connected properly before switching on the 230V AC mains supply.

Working

The pump must start (when the switch S6 is closed and MCB is in 'on' position). The pump must stop when the OHT water level reaches the set maxi-

mum level.

The reset push-to-off switch S1 is pressed to reset the complete circuit and alarm when auto timer cut-off takes place. This button can also be used to start the pump to top-up the OHT any time you desire. The pump stops automatically when the OHT is filled to the maximum level.

The by-pass switch (MCB) facilitates you to manually operate the pump starter when the water level controller circuit is not working properly. A 20-ampere MCB can be used safely to handle large current surges encountered during starting of the submersible pump motor. Any other switch that can handle the peak current requirement of the pump motor can be used for this purpose. ●

The author retired as junior technical officer from DGDA



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